

David A. Plappert

Principal Cloud Architect · Enterprise Architecture · AWS Reliability Lead

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US Patent 11,146,689, Cloud Architecture (2021) · 2x AWS City on a Cloud award (2016, 2018) · 19 years on AWS, since EC2 beta

SUMMARY

- Principal-level cloud architect with 19 years building and operating mission-critical AWS SaaS platforms
- Patent holder; two-time AWS innovation award winner; 18F (federal digital services, GSA) advisor while at IdeaCrew
- Ships AI-augmented engineering tooling in production today: Claude Code orchestrated CI/CD, agentic deployment workflows, MCP-based tool integration, multi-agent parallel orchestration
- Currently the Caterpillar-side AWS reliability lead for **Vertex 3D**, Caterpillar's vendor-built real-time 3D collaboration platform powering aftermarket commerce
- Specializes in AWS CDK gatekeeping, distributed systems reliability, observability, and standing up production-critical workloads
- **Industry depth:** Insurance, Payments, and Finance (10 yrs at IdeaCrew on state health insurance exchanges, \$500M/yr atomic payments, IRS / Stripe / EDI integrations); Manufacturing (Caterpillar aftermarket-commerce backbone); Healthcare and Federal Government (10 yrs DCHBX, MA HCB, CoverME, DC DBH, 18F advisor); Retail and Point-of-Sale (Best Buy, billion-dollar BestBuy.com and POS systems on AWS); Media / Streaming (PBS RTMP video platform; production RTMP and SRT live ingest pipelines)

PROFESSIONAL EXPERIENCE

Caterpillar Inc. (via TEKsystems), Lead Cloud Engineer / AWS Reliability Lead

September 2025 – Present · Chillicothe, IL

- AWS reliability lead for **Vertex 3D**, Caterpillar's real-time 3D collaboration platform powering aftermarket commerce, ~\$3,000/minute in revenue when running, ~30 downstream publisher integrations
- Built and operate multiple production applications on Vertex 3D, direct hands-on experience with the platform's APIs, SDKs, and integration surface as a power consumer
- Sole Caterpillar-side reviewer of monthly vendor-shipped AWS CDK releases; extensive pre-deploy testing and architectural review to catch issues before production
- Designed and own load testing, monitoring, and alerting strategy across the Vertex 3D production deployment
- Built and operate a Claude Code agentic workflow that runs production deployments, pre-production dry-runs, automated bug detection, ticket filing, and end-to-end CI/CD pipeline orchestration; agents read architecture and open work, propose changes, file tickets, push code, and watch the deploy
- Parallel multi-agent orchestration against several stacks simultaneously, bridged into GitHub, AWS CLI, and the team's ticketing system through custom MCP servers and tool definitions I wrote
- Cross-team escalation point across both AWS and Node.js stacks for Caterpillar's Publisher Distribution Team (PDT), scope well beyond the original Node.js Developer SOW
- Technical lead for Caterpillar's Node.js workstream on Vertex 3D; actively mentoring junior Node.js engineer

- Standing AWS authority on Vertex 3D production incidents; no other Caterpillar-side AWS team coverage of this workload since deployment
- **Stack:** AWS (Fargate, ECS, S3, RDS / Aurora Serverless, Lambda, CloudWatch), AWS CDK, CloudFormation, Node.js, Java / JVM, GitHub Actions, Claude Code, Anthropic SDK, MCP

IdeaCrew, Principal Cloud Architect & DevOps Lead

2015 – 2025 · Remote

- Employee #3; led cloud architecture and DevOps as the company grew past 100 employees
- Managed \$1M/month cloud spend across 12 AWS and 3 Azure accounts supporting client engagements worth >\$50M annually
- Owned the AWS architecture that won **2x AWS City on a Cloud awards (2016, 2018)** on the DC Health Benefit Exchange platform
- Final escalation point for production incidents across every client engagement, company-wide, backed by extensive **New Relic** observability (APM, transaction tracing, alerting)
- Owned full tech stack and AWS infrastructure for DC Health Benefit Exchange (DCHBX), Massachusetts Health Connector for Business, CoverME (Maine Health Exchange), and DC Department of Behavioral Health
- Led multi-year state health insurance migrations from federal FedHub to state-based exchanges, worth tens of millions; team held the fastest transition record off FedHub
- **18F (federal digital services, GSA) advisor**, engaged by multiple 18F chapters in 2018 for AWS architecture consultation on state-government health exchange replication
- Designed distributed data consistency patterns across multi-region state-exchange architecture, idempotent EDI processing, eventual consistency reconciliation, and atomic payment workflows handling >\$500M annually under HIPAA, SOC 2, and IRS Publication 1075 compliance
- Led SOC 2 certification and annual audit processes for state engagements; same engineering discipline transfers cleanly to IP protection, license scanning, secret prevention, and audit trail design for AI-assisted SDLC
- Led monolith-to-microservices migration onto AWS EKS, ECS, and Lambda
- Solo-built data warehouse spanning 8 databases (MongoDB BI Connector, MySQL, PostgreSQL) feeding Tableau; drove \$1M annual contract increase
- Developed 300+ automated jobs and scripts enabling HIPAA- and SOC 2-compliant data flows; drove an additional \$2M annual contract increase
- Led SOC 2 certification and annual audit processes for state engagements
- Delivered all client engagements through agile methodology, scrum sprints, CI/CD pipelines, code reviews, and continuous integration practices
- Integrated enterprise platforms across Slack, Twilio (voice/SMS), IRS, Stripe, Tableau, SugarCRM, AWS, Azure, MySQL, PostgreSQL, Jenkins, MongoDB, Oracle, GitHub Actions
- **Stack:** Ruby on Rails, Node.js, Angular, AWS, Azure, EKS, ECS, Docker, Terraform, MongoDB, MySQL (Aurora Serverless), DynamoDB, PostgreSQL, Oracle

PBS, Cloud Engineer

2014

- Built and operated AWS-based RTMP video streaming platform with 99.9% uptime
- Implemented adaptive bandwidth modulation, increasing viewer watch time by 158%
- Managed a development team of two direct reports

Argonne National Laboratory, Senior Cloud Architect

2013

- Designed and implemented a PHP web application for federal grant applications
- Streamlined grant workflows, driving a 283% increase in lab revenue

Best Buy, Senior Cloud DevOps

2012

- Owned cloud migration of **BestBuy.com and Point-of-Sale, billion-dollar revenue systems**, plus New Product Ingestion, from on-premises infrastructure to AWS; automated scaling and patching
- Built production services in **Ruby on Rails** and operated **Java / JVM** workloads across the BestBuy.com and Point-of-Sale stacks, with Cassandra in the BestBuy.com data layer
- Reduced infrastructure costs by over 50% while improving deployment speed
- Led a 6-member DevOps engineering team

SELECTED CREDENTIALS

- **US Patent 11,146,689**, Cloud Architecture (Issued October 12, 2021)
- **AWS City on a Cloud Award**, 2016 and 2018
- **19 years on AWS**, operating production workloads since EC2 beta
- **18F advisor** (federal digital services / GSA), AWS architecture consultation during IdeaCrew tenure

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (Fargate, ECS, EKS, S3, RDS, Lambda, CloudWatch, CDK, CloudFormation), Azure, GCP
- **AI Engineering & Agent Tooling:** Claude Code CLI for production engineering work, Anthropic API and SDK integration, multi-agent parallel orchestration, context engineering and prompt design, persistent memory and documentation organization for AI consumption, MCP servers and custom tool definitions, AI-assisted CDK / infrastructure development, evaluation harnesses, RAG patterns, model and cost / latency trade-offs
- **CI / CD & Developer Productivity:** GitHub Actions, Jenkins, AWS CodePipeline, Chef, agentic CI/CD orchestration, automated PR review and test synthesis, flaky test triage, agent-assisted incident analysis and runbook automation
- **AI-SDLC Governance:** IP protection patterns, secret scanning, open-source license compliance, audit trail design (transferred from SOC 2 / HIPAA / IRS Pub 1075 background)
- **Infrastructure as Code & Containers:** AWS CDK, CloudFormation, Terraform, Docker, Kubernetes, EKS
- **Languages:** Ruby, Node.js, JavaScript / TypeScript, Python, Bash, SQL, PHP (author); Java / JVM (operate on AWS)
- **Frameworks:** Ruby on Rails, Nuxt / Vue.js, Next.js / React, Astro, AWS CDK, Express.js, Angular, Serverless Framework
- **Databases:** PostgreSQL, MySQL (Aurora Serverless, TiDB, MariaDB), DynamoDB, Cassandra, MongoDB (MongoBI), Oracle, Redis
- **Video & Streaming:** RTMP and SRT live ingest, adaptive bitrate delivery, low-latency streaming pipelines on AWS

- **Observability:** New Relic (APM, transaction tracing, alerting), CloudWatch, OpenTelemetry, structured logging, distributed tracing
- **Architecture:** Distributed systems, event-driven architecture, microservices, EDI systems, multi-region, hybrid cloud
- **Compliance & Security:** HIPAA, SOC 2 certification and audit, IRS Publication 1075, security automation
- **Enterprise Integrations:** Slack API, Twilio (voice / SMS), Stripe, Tableau, EDI, IRS, REST APIs